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e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC ASSESSMENT TEST – 09** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If $2x^2 + px - 15 = 0$ has -5 as a root, and $p(x^2 + x) + k = 0$ has equal roots, then $k =$: [1]

(A) $\frac{7}{4}$ **(B)** $\frac{4}{7}$

(C) $-\frac{7}{4}$ **(D)** 7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{7}{4}$

Since -5 is a root of $2x^2 + px - 15 = 0$, substituting $x = -5$:

$$2(-5)^2 + p(-5) - 15 = 0 \implies 50 - 5p - 15 = 0 \implies 35 - 5p = 0 \implies p = 7$$

Substituting $p = 7$ into the second equation gives $7x^2 + 7x + k = 0$ $(a = 7, b = 7, c = k)$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

$$(7)^2 - 4(7)(k) = 0 \implies 49 = 28k \implies k = \frac{49}{28} = \boxed{\frac{7}{4}}$$

Hence, option **(A)** is correct. **[1 mark]**

Q2. If $x = 2$ and $x = 3$ are the roots of the equation $3x^2 - 2kx + 2m = 0$, then: [1]

(A) $k = \frac{15}{2},\ m = 9$ **(B)** $k = 9,\ m = \frac{15}{2}$

(C) $k = 15,\ m = 18$ **(D)** $k = -15,\ m = -9$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $k = \frac{15}{2},\ m = 9$

Comparing $3x^2 - 2kx + 2m = 0$ with $ax^2 + bx + c = 0$, we have $a = 3$, $b = -2k$, $c = 2m$, and roots $\alpha = 2$, $\beta = 3$.

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

- Sum of roots: $2 + 3 = -\frac{-2k}{3} \implies 5 = \frac{2k}{3} \implies k = \frac{15}{2}$.
- Product of roots: $2 \times 3 = \frac{2m}{3} \implies 6 = \frac{2m}{3} \implies m = 9$.

Thus $\boxed{k = \frac{15}{2},\ m = 9}$. Hence, option **(A)** is correct. **[1 mark]**

Q3. The value of k for which $x = -2$ is a root of the equation $3x^2 + 4x + 2k = 0$ is: [1]

(A) 2 **(B)** -2

(C) 4 **(D)** -4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** -2

Since $x = -2$ is a root of $p(x) = 3x^2 + 4x + 2k = 0$, we must have $p(-2) = 0$.

Formula Used: If $x = \alpha$ is a root of $p(x) = 0$, then $p(\alpha) = 0$

Substituting $x = -2$:

$$3(-2)^2 + 4(-2) + 2k = 0 \implies 12 - 8 + 2k = 0 \implies 4 + 2k = 0 \implies \boxed{k = -2}$$

Hence, option **(B)** is correct. **[1 mark]**

Q4. If the sum of the roots of a quadratic equation is 6 and the product of roots is 8, the equation is: [1]

(A) $x^2 - 6x + 8 = 0$ **(B)** $x^2 + 6x + 8 = 0$

(C) $x^2 - 6x - 8 = 0$ **(D)** $x^2 + 8x - 6 = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $x^2 - 6x + 8 = 0$

Given sum of roots $S = \alpha + \beta = 6$ and product of roots $P = \alpha\beta = 8$.

Formula Used: Quadratic equation with given roots: $x^2 - Sx + P = 0$

Substituting $S = 6$ and $P = 8$:

$$\boxed{x^2 - 6x + 8 = 0}$$

Hence, option **(A)** is correct. **[1 mark]**

Q5. If the roots of the quadratic equation $kx^2 - 2\sqrt{5}x + 4 = 0$ are equal, then $k =$: [1]

(A) $\frac{5}{4}$ **(B)** $\frac{4}{5}$

(C) 5 **(D)** $\frac{\sqrt{5}}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{5}{4}$

Comparing $kx^2 - 2\sqrt{5}x + 4 = 0$ $(k \neq 0)$ with $ax^2 + bx + c = 0$, we have $a = k$, $b = -2\sqrt{5}$, and $c = 4$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

Setting $D = 0$:

$$(-2\sqrt{5})^2 - 4(k)(4) = 0 \implies 4(5) - 16k = 0 \implies 20 = 16k \implies k = \frac{20}{16} = \boxed{\frac{5}{4}}$$

Hence, option **(A)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): The sum of the roots of the quadratic equation $3x^2 - 9x + 5 = 0$ is 3.

Reason (R): For any quadratic equation $ax^2 + bx + c = 0$, the sum of roots is given by $-\frac{b}{a}$.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

For $3x^2 - 9x + 5 = 0$, $D = (-9)^2 - 4(3)(5) = 81 - 60 = 21 > 0$ (real roots exist), and $a = 3$, $b = -9$.

Formula Used: Sum of roots $(\alpha + \beta) = -\frac{b}{a}$

$$\alpha + \beta = -\frac{-9}{3} = 3$$

Thus Assertion (A) is **true**, Reason (R) states the exact general formula and is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

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Q7. If one root of the quadratic equation $3x^2 - 10x + k = 0$ is reciprocal of the other, find the value of k. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the roots of $3x^2 - 10x + k = 0$ be α and $\frac{1}{\alpha}$ $(\alpha \neq 0)$.

Comparing with $ax^2 + bx + c = 0$, we have $a = 3$, $b = -10$, and $c = k$.

Formula Used: Product of roots $(\alpha \cdot \beta) = \frac{c}{a}$

Applying the product of roots formula:

$$\alpha \cdot \frac{1}{\alpha} = \frac{k}{3} \implies 1 = \frac{k}{3} \implies \boxed{k = 3} \quad \textbf{[2 marks]}$$

(Verification: At $k = 3$, $3x^2 - 10x + 3 = (3x - 1)(x - 3) = 0$ has reciprocal roots 3 and $\frac{1}{3}$.)

Q8. Solve the following quadratic equation for x by factorization: $4x^2 - 4ax + (a^2 - b^2) = 0$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given equation: $4x^2 - 4ax + (a^2 - b^2) = 0$. Using $a^2 - b^2 = (a - b)(a + b)$:

Formula Used: $A^2 - B^2 = (A - B)(A + B)$

Notice that $-2(a - b) - 2(a + b) = -4a$ and $[-2(a - b)][-2(a + b)] = 4(a^2 - b^2)$.

Splitting the middle term:

$$\begin{aligned} 4x^2 - 2(a - b)x - 2(a + b)x + (a - b)(a + b) &= 0 \\ 2x[2x - (a - b)] - (a + b)[2x - (a - b)] &= 0 \quad \textbf{[1 mark]} \\ [2x - (a - b)][2x - (a + b)] &= 0 \end{aligned}$$

Equating each factor to zero:

$$2x = a - b \implies x = \frac{a - b}{2} \quad \text{or} \quad 2x = a + b \implies x = \frac{a + b}{2}$$

Hence, the roots are $\boxed{x = \frac{a + b}{2} \text{ and } x = \frac{a - b}{2}}$. **[1 mark]**

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Q9. Solve the following equation for x: $\dfrac{4}{x} - 3 = \dfrac{5}{2x+3}, \quad x \neq 0, -\dfrac{3}{2}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given equation: $\dfrac{4}{x} - 3 = \dfrac{5}{2x+3}$, where $x \neq 0, -\dfrac{3}{2}$.

Formula Used: $\dfrac{4-3x}{x} = \dfrac{5}{2x+3}$

Taking the LCM on the LHS and cross-multiplying:

$$(4-3x)(2x+3) = 5x$$
$$8x + 12 - 6x^2 - 9x = 5x \quad \textbf{[1 mark]}$$
$$-6x^2 - x + 12 = 5x \implies 6x^2 + 6x - 12 = 0 \implies x^2 + x - 2 = 0 \quad \text{(dividing by 6)} \quad \textbf{[1 mark]}$$

Factorising ($2 \times (-1) = -2$ and $2 - 1 = 1$):

$$x^2 + 2x - x - 2 = 0 \implies x(x+2) - 1(x+2) = 0 \implies (x+2)(x-1) = 0 \implies x = -2 \text{ or } x = 1$$

Both roots satisfy $x \neq 0, -\frac{3}{2}$. Hence, the solutions are $\boxed{x = 1 \text{ and } x = -2}$. **[1 mark]**

Q10. If the quadratic equation $(1+m^2)x^2 + 2mcx + (c^2 - a^2) = 0$ has equal roots, prove that $c^2 = a^2(1+m^2)$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $(1+m^2)x^2 + 2mcx + (c^2 - a^2) = 0$ with $Ax^2 + Bx + C = 0$, we identify $A = 1 + m^2 \neq 0$, $B = 2mc$, and $C = c^2 - a^2$.

Formula Used: Equal real roots $\iff D = B^2 - 4AC = 0$

Setting the discriminant $D = 0$:

$$(2mc)^2 - 4(1+m^2)(c^2 - a^2) = 0 \quad \textbf{[1 mark]}$$
$$4m^2c^2 - 4(c^2 - a^2 + m^2c^2 - m^2a^2) = 0$$
$$m^2c^2 - c^2 + a^2 - m^2c^2 + m^2a^2 = 0 \quad \text{(dividing by 4)} \quad \textbf{[1 mark]}$$

Cancelling $m^2c^2 - m^2c^2 = 0$:

$$-c^2 + a^2 + m^2a^2 = 0 \implies c^2 = a^2 + a^2m^2 \implies \boxed{c^2 = a^2(1+m^2)} \quad \text{(Hence Proved)} \quad \textbf{[1 mark]}$$

Q11. In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of her marks would have been 210. Find her marks in the two subjects. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let Shefali's marks in Mathematics be x. Then her marks in English $= 30 - x$ (since total $= 30$).

Formula Used: New Maths marks $= x + 2$, New English marks $= (30 - x) - 3 = 27 - x$

According to the given condition:

$$(x + 2)(27 - x) = 210$$
$$27x - x^2 + 54 - 2x = 210 \quad \textbf{[1 mark]}$$
$$-x^2 + 25x + 54 - 210 = 0 \implies -x^2 + 25x - 156 = 0 \implies x^2 - 25x + 156 = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Factorising ($-12 \times (-13) = 156$ and $-12 - 13 = -25$):

$$x^2 - 12x - 13x + 156 = 0 \implies x(x - 12) - 13(x - 12) = 0$$
$$\implies (x - 12)(x - 13) = 0 \implies x = 12 \text{ or } x = 13 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

- If Marks in Mathematics $= 12$, then Marks in English $= 30 - 12 = 18$. (Check: $14 \times 15 = 210$.)
- If Marks in Mathematics $= 13$, then Marks in English $= 30 - 13 = 17$. (Check: $15 \times 14 = 210$.)

Hence, $\boxed{\text{Maths} = 12,\ \text{English} = 18}$ OR $\boxed{\text{Maths} = 13,\ \text{English} = 17}$. **[1 mark]**

Q12. A piece of cloth costs Rs. 200. If the piece were 5 m longer and each metre of cloth cost Rs. 2 less, the total cost of the piece would have remained unchanged. How long is the piece and what is its original rate per metre? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the original length of the cloth piece be x m ($x > 0$).

Since total cost is Rs. 200, original rate per metre $=$ Rs. $\dfrac{200}{x}$.

Formula Used: Total Cost = (Length) $\times$ (Rate per metre), $(x + 5)\left(\dfrac{200}{x} - 2\right) = 200$

Setting up the equation with new length $(x + 5)$ m and reduced rate $\left(\dfrac{200}{x} - 2\right)$:

$$(x + 5)\left(\frac{200 - 2x}{x}\right) = 200$$
$$(x + 5)(200 - 2x) = 200x \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$
$$200x - 2x^2 + 1000 - 10x = 200x$$
$$-2x^2 - 10x + 1000 = 0 \implies x^2 + 5x - 500 = 0 \quad \text{(dividing by } -2\text{)} \quad \textbf{[1 mark]}$$

Factorising ($25 \times (-20) = -500$ and $25 - 20 = 5$):

$$x^2 + 25x - 20x - 500 = 0 \implies x(x + 25) - 20(x + 25) = 0 \implies (x + 25)(x - 20) = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Thus $x = -25$ or $x = 20$. Since length cannot be negative ($x > 0$), we reject $x = -25$.

Therefore, length of cloth $= 20$ m, and original rate per metre $= \dfrac{200}{20} =$ Rs. 10 per metre.

Hence, $\boxed{\text{Length} = 20 \text{ m and Original rate} = \text{Rs. } 10 \text{ per metre}}$. **[1 mark]**

Q13. Case Study: Sports Biomechanics & Cricket Bowling Pitch Trajectory

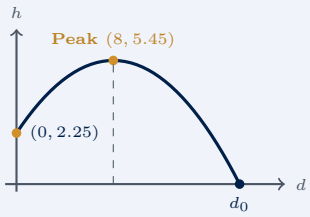
During a high-performance cricket biomechanics camp, the vertical flight trajectory of a delivery bowled by a fast bowler is captured by ultra-high-speed motion cameras. The height h (in metres) of the ball above the pitch surface at a horizontal distance d (in metres) from the bowling release point is modeled by $h(d) = -0.05d^2 + 0.8d + 2.25$. [4]

(i) At what initial release height ($d = 0$) did the bowler release the ball from his hand? **[1]**

(ii) Set up the quadratic equation in d in standard integer form to find when the ball hits the pitch ($h = 0$). **[1]**

(iii) Solve the quadratic equation to determine the horizontal distance d at which the ball pitches. **OR** Determine the maximum height attained by the cricket ball above the pitch during its flight. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]



As plotted alongside, the cricket ball is released at $d = 0$ m at height 2.25 m, rises along the parabolic trajectory $h(d) = -0.05d^2 + 0.8d + 2.25$ to a peak height at $d = 8$ m, and finally hits the pitch ($h = 0$) at $d \approx 18.44$ m.

Formula Used: $h(d) = -0.05d^2 + 0.8d + 2.25, \qquad d = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(i) Initial Release Height ($d = 0$): Substituting $d = 0$ into $h(d)$:

$$h(0) = -0.05(0)^2 + 0.8(0) + 2.25 = \boxed{2.25 \text{ m}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Integer Form: When the ball hits the pitch, $h(d) = 0$:

$$-0.05d^2 + 0.8d + 2.25 = 0 \xrightarrow{\times(-20)} \boxed{d^2 - 16d - 45 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Pitching Distance d): Solving $d^2 - 16d - 45 = 0$ using the quadratic formula ($a = 1, b = -16, c = -45$):

$$d = \frac{16 \pm \sqrt{(-16)^2 - 4(1)(-45)}}{2(1)} = \frac{16 \pm \sqrt{256 + 180}}{2} = \frac{16 \pm \sqrt{436}}{2} = \frac{16 \pm 2\sqrt{109}}{2} = 8 \pm \sqrt{109}$$

Since forward horizontal distance $d > 0$ and $8 - \sqrt{109} \approx -2.44 < 0$, we reject $8 - \sqrt{109}$.
Hence, pitching distance $= \boxed{(8 + \sqrt{109}) \text{ m} \approx 18.44 \text{ m}}$. **[2 marks]**

OR (Alternative Part — Maximum Height via Completing the Square): Rewriting $h(d)$:

$$\begin{aligned} h(d) &= -0.05(d^2 - 16d) + 2.25 \\ &= -0.05\left[d^2 - 2(d)(8) + 8^2 - 8^2\right] + 2.25 \\ &= -0.05(d - 8)^2 + (0.05 \times 64) + 2.25 = -0.05(d - 8)^2 + 3.20 + 2.25 = -0.05(d - 8)^2 + 5.45 \end{aligned}$$

Since $-0.05(d - 8)^2 \leq 0$ for all real d, the maximum occurs when $d = 8$ m, giving $\boxed{\text{Maximum height} = 5.45 \text{ m}}$. **[2 marks]**

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